

Atrium Radiant Heating System

California Polytechnic State University – San Luis Obispo
Department of Mechanical Engineering

ME 448: Thermal System Design
Dr. Christopher Pascual

Isaac Golan & Julian Rubinstein

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Introduction

A new community center is being built in Atascadero, California. The entrance of the community center will feature a two-story atrium used for special events. The concrete foundation of the atrium will be 20 ft wide by 40 ft long (for a total area of 800 ft²). The owner wants a new radiant heating system installed in the atrium. The radiant heating system will pump heated water through tubes embedded in the concrete foundation. As a result, the concrete will heat up and transfer the heat to the atrium air space (Figure 1).

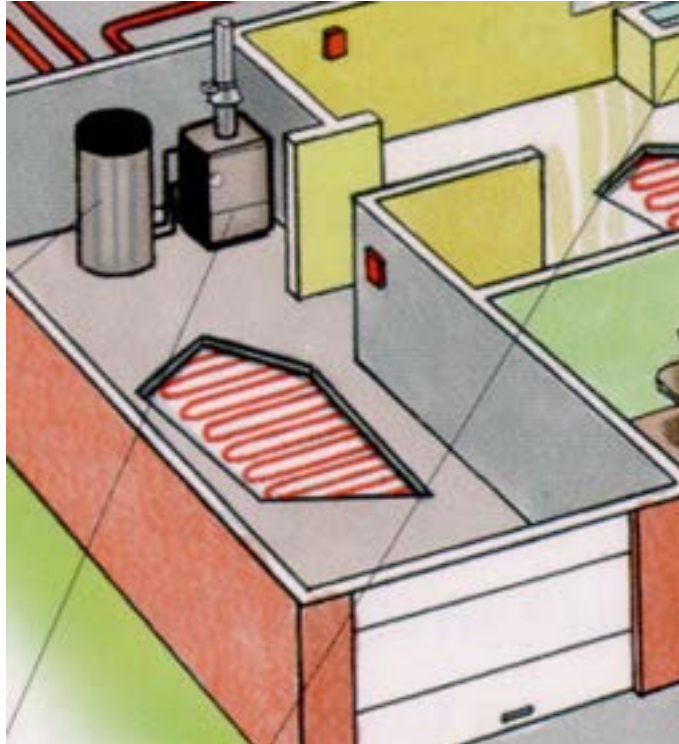


Figure 1. Embedded Tubes in Concrete for Garage Heating (similar to atrium system)

The system will use natural convection and radiation to heat the atrium, transferring energy through a network of tubes within the concrete slab. The heat transfer modeling, material selection, tube sizing, tube spacing, pump selection, and optimization of system components are detailed in this report. Steady state and transient finite difference conduction models and EES simulations were employed to predict system performance under various environmental conditions. Finally, economic analysis and system control strategies were conducted to ensure the feasibility and practicality of the proposed design.

Heat Load

According to weather data provided by the Atascadero Chamber of Commerce, Atascadero has a mean daily minimum temperature of 31.9°F in December. Table 1 shows the complete temperature data for the year in Atascadero.

Table SEQ Table \# ARABIC 1. Temperature Data for Atascadero, CA.				
Month	Mean Daily Max	Mean Daily Min	Record High	Record Low
Jan	60.7	32.8	79	11
Feb	65	36	83	18
Mar	68	37.2	91	20
Apr	73.6	39.5	98	25
May	79.9	43.2	103	30
Jun	87	47.3	111	31
Jul	93.9	50.1	113	37
Aug	93.6	49.6	108	37
Sep	89.9	46.5	111	33
Oct	81.6	40.9	105	19
Nov	69.2	35.4	95	14
Dec	61.7	31.9	87	15

Design assumptions are shown in Table 2.

Table 2. Design Assumptions

Design Day Temperature	30°F
Indoor Set Point Temperature	68°F
Window Height	30 ft
Window Area	1200 ft ²
Concrete Surface Area	800 ft ²
Emissivity of All Surfaces	0.9

Using EES, the model accounted for conduction resistance through the window, natural convection, and radiation exchange between the concrete surface and the window.

From these calculations, the design heat load was determined to be $q_{\text{load}} = 13224$ Btu/hr. At the design heat load, the model determined the average concrete surface temperature: $T_s = 77.2^\circ\text{F}$.

The following parametric table (Table 3) shows the concrete surface temperature (T_s) and heat load (q_{load}) at 10 outside temperatures (T_{outside}) between 30°F to 60°F .

Table 3. Parametric Table showing the concrete surface temperature and heat load response to varying outside temperature.

$T_{\text{outside}} (^\circ\text{F})$	$T_s (^\circ\text{F})$	$q_{\text{load}} (\text{Btu/hr})$
30	77.2	13224
33.3	76.5	12064
36.6	75.7	10904
40	75	9744
43.3	74.2	8584
46.6	73.4	7424
50	72.6	6264
53.3	71.8	5104
56.6	71	3944
60	70.2	2784

From the parametric table, Figure 2 and Figure 3 show the variation in concrete surface temperature (T_s) and variation in heat load (q_{load}) as a function of outside temperature (T_{outside}).

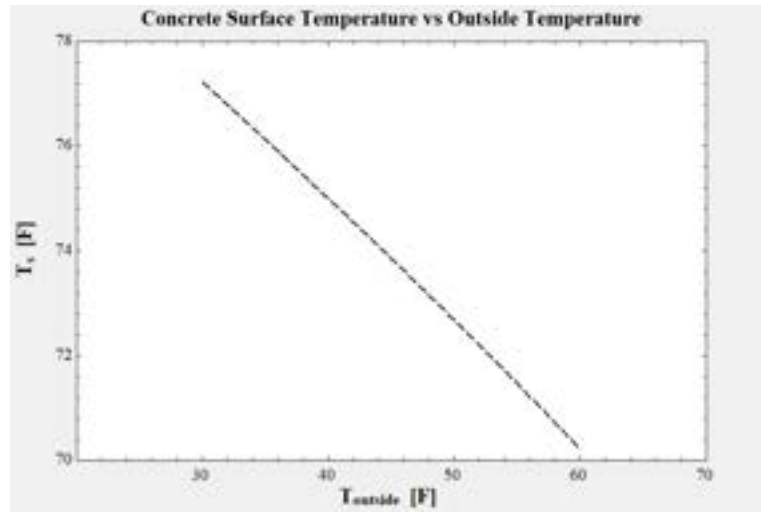


Figure 2. Concrete Surface Temperature as a function of Outside Temperature. The heating required decreases with increasing temperature. Thus, since the concrete surface is the source of the radiant heating, its temperature must be the highest when outside temperature is the lowest, and the lowest when outside temperature is the highest.

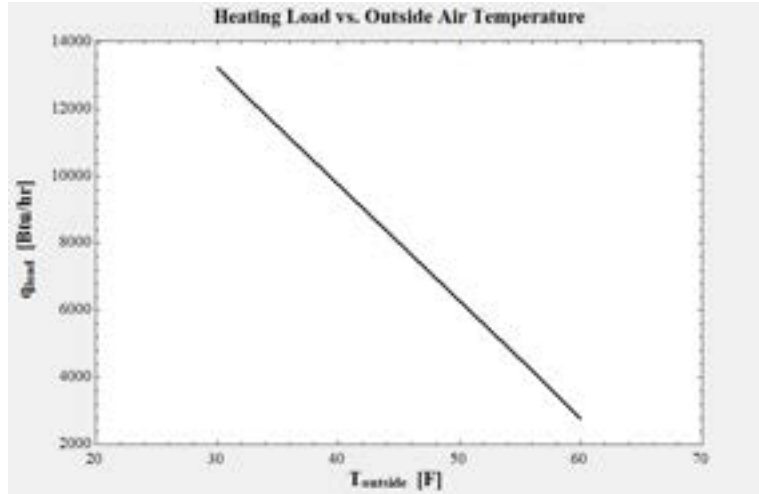


Figure 3. Heating Load as a function of Outside Temperature. Similar to the concrete surface temperature, the heating required decreases with increasing temperature. Thus, the heating load must be the highest when outside temperature is the lowest, and the lowest when outside temperature is the highest.

The EES model determined the total heat transfer coefficient: $h_{\text{total}} = 1.79 \text{ Btu/hr-ft}^2\text{-}^\circ\text{F}$, or $10.17 \text{ W/m}^2\text{-}^\circ\text{C}$.

This value was compared to the model presented in Cholewa et al. (2013), which accounts for radiation and convection when determining total heat transfer coefficients for radiant floor heating and cooling systems.

Figure 4 shows the measured total heat transfer coefficient depending on temperature difference for a heated radiant floor in Cholewa, et al. This is compared to the total heat transfer coefficient determined from the EES model, superimposed on the figure in **red**.

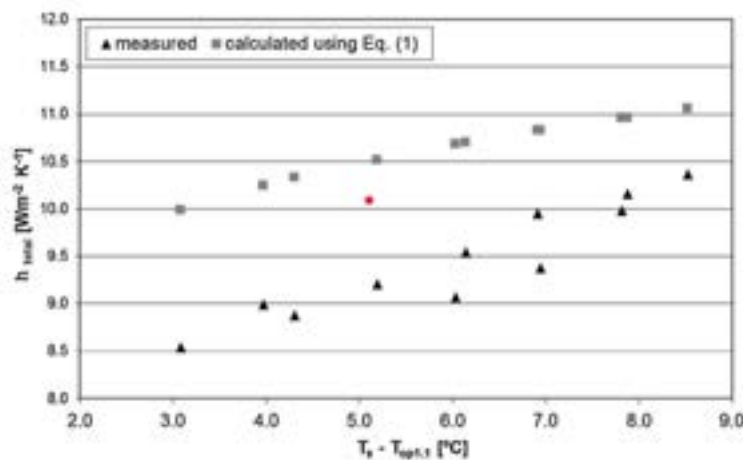


Figure 4. Total Heat Transfer Coefficient (Cholewa et al.) compared to EES model.

The study assumes that commonly used heat transfer coefficients for radiant floor heating and cooling are overestimated and should be experimentally verified. This is evident in Figure 4. The EES model is only slightly closer to the measured results compared to Cholewa's calculated results. However, there are critical similarities and differences in the assumptions and experimental conditions of these two studies.

Both models analyze radiant floor heating systems by considering heat transfer through convection and radiation. They use established natural convection correlations from standard heat transfer textbooks and treat radiant surfaces as diffuse gray, with an emissivity of 0.9. Both models also determine the surface temperature of the heated floor to maintain indoor comfort and analyze how it varies with outdoor conditions.

Despite these similarities, the models differ significantly in their scope and application. The Cholewa et al. model is a controlled laboratory experiment, aimed at refining heat transfer coefficients for radiant floors, while the project model applies these findings to a large-scale atrium heating system. A key distinction is that Cholewa et al. does not explicitly model ground temperature or insulation, whereas the project model assumes a constant ground temperature of 55°F and incorporates insulation layers in its design.

Steady State 2D Conduction Model

The steady state 2D conduction model consists of a grid of finite difference equations (FDEs) that calculate the temperature of each node given the temperatures of the surrounding nodes. These equations were derived by solving the energy balance equations for the 9 types of nodes, listed below. Notably, boundary nodes have half of the area of interior nodes, and boundaries with heat transfer include the appropriate heat transfer equations in their FDEs.

Upper-Left Corner Node:
$$\frac{T_{right} + T_{below} + \frac{h_{total}\Delta x}{k_c} T_{\infty}}{2 + \frac{h_{total}\Delta x}{k_c}}$$

Upper-Middle Nodes:
$$\frac{T_{left} + T_{right} + 2T_{below} + 2\frac{h_{total}\Delta x}{k_c} T_{\infty}}{4 + 2\frac{h_{total}\Delta x}{k_c}}$$

Upper-Right Corner Node:
$$\frac{T_{left} + T_{below} + \frac{h_{total}\Delta x}{k_c} T_{\infty}}{2 + \frac{h_{total}\Delta x}{k_c}}$$

Left Wall Nodes:
$$\frac{T_{above} + 2T_{right} + T_{below}}{4}$$

Middle Nodes:
$$\frac{T_{above} + T_{below} + T_{left} + T_{right}}{4}$$

Right Wall Nodes:
$$\frac{T_{above} + 2T_{left} + T_{below}}{4}$$

Upper- Left Insulation Node:

$$\frac{k_c}{2(k_c + k_l)}(T_{above} + T_{right}) + \frac{k_l}{2(k_c + k_l)}(T_{below} + T_{right})$$

Upper-Middle Insulation Nodes:

$$\frac{k_c}{4(k_c + k_l)}(T_{left} + 2T_{above} + T_{right}) + \frac{k_l}{4(k_c + k_l)}(T_{left} + 2T_{below} + T_{right})$$

Upper-Right Insulation Node:

Δx	0.1 in
h_{total}	$1.79 \frac{Btu}{hr ft^2 \text{ } ^\circ F}$
$\overline{q''}_s$	$16.83 \frac{Btu}{hr ft^2}$
q_s	$13492 \frac{Btu}{hr}$
T_s	$77.4 \text{ } ^\circ F$

$$\frac{k_c}{2(k_c + k_l)}(T_{above} + T_{left}) + \frac{k_l}{2(k_c + k_l)}(T_{below} + T_{left})$$

Table 4: Excel model parameters and results

We used a Δx of 0.1 in for our 2D steady-state model, since it resulted in a good balance of accuracy and computation speed. This value was also used for Δy for simplicity. After running the steady-state model, we found that the overall heat transfer coefficient was 1.79 Btu/hr ft²°F. The average surface heat flux was 16.83 Btu/hr ft², resulting in a total surface heat flux of 13492

Btu/hr for the entire atrium floor. This heat flux is very close to our calculated heat load of 13224 Btu/hr. The average surface temperature was 77.4 °F, which matched very closely with our EES model.

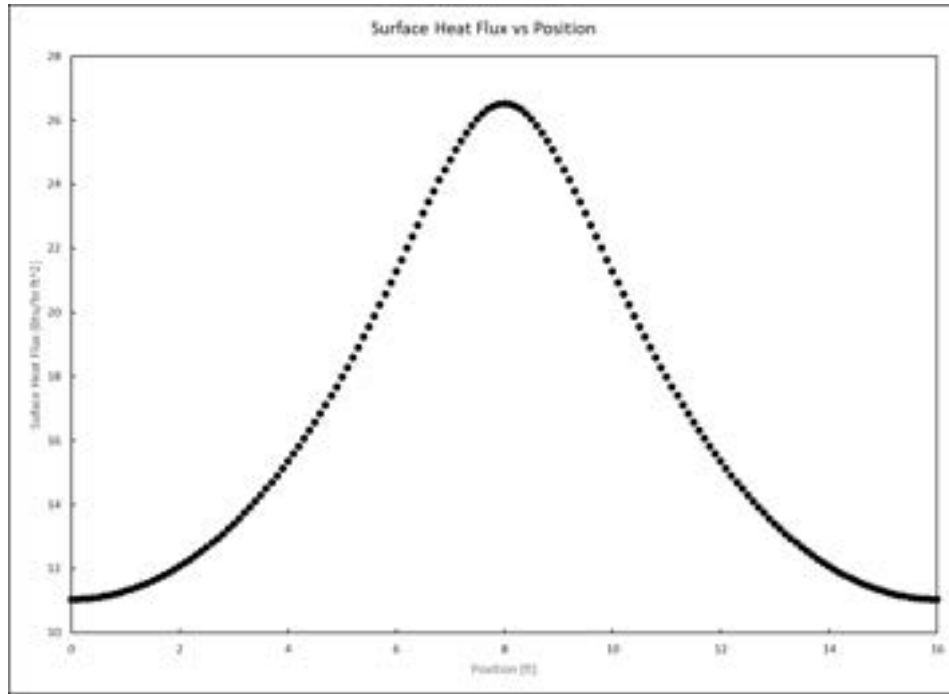


Figure 5. Concrete surface heat flux plotted as a function of position.

The above figure shows the surface heat flux of each surface node plotted as a function of position. There is a clear peak at the center, where the heating pipe is located, and the change in surface heat flux approaches zero at the edges. This matches our expectations, as the surface above the heating pipe is the hottest area, and the sides of the nodal network are adiabatic.

The surface temperature found using the EES model was 77.28 °F, which is very similar to the temperature found using the Excel model, which was 77.4 °F. The Excel model resulted in a slightly higher total heat flux than the EES model, which would account for the slightly higher surface temperature. The EES model calculates the heat flux based on the average surface temperature, so it would make sense that the two models agree on this value.

We chose to use two layers of 1-in thick insulation to minimize heat loss to the ground. Then, we found a reasonable tube surface temperature to achieve the required heat flux from the surface of the concrete. The total heat loss to the ground is 2513.9 Btu/hr.

$$R_{th, con} = (T_{tube} - T_s) \left(\frac{A_s}{q_s} \right) = 0.866 \frac{hr ft^2 \text{ } ^\circ F}{Btu}$$

$$R_{th, ins} = (T_{tube} - T_{ground}) \left(\frac{A_s}{q_{ground}} \right) = 11.77 \frac{hr ft^2 \text{ } ^\circ F}{Btu}$$

$$T_{tube} = \frac{R_{th,ins}}{\left(\frac{A_s}{q_{ground}}\right)} + T_{ground} = \frac{R_{th,c}}{\left(\frac{A_s}{q_s}\right)} + T_s = 92^{\circ}F$$

The tube spacing was chosen by testing different values until the desired heat transfer rate was achieved at a reasonable tube surface temperature. This was done to reduce the temperature requirement for the heat exchanger. We chose to use a tube spacing of 16 inches, which drove the decision to use 3/4-inch PEX tubing, since it has a minimum bending diameter of 14 inches. This is important for minimizing cost, as it eliminates the need to purchase fittings.

Transient 2D Conduction Analysis

Like the steady state 2D conduction analysis, the transient conduction model consists of a grid of finite difference equations (FDEs) that calculate the temperature of each node given the temperatures of the surrounding nodes. However, each node is determined implicitly instead of explicitly. These equations were derived by solving the energy balance equations for the 12 types of nodes, listed below.

Upper-Left Corner Node:
$$\frac{1}{2+Bi+\frac{1}{2Fo_c}} (T_{right} + T_{below} + \frac{1}{2Fo_c} T_{node}^P + BiT_{\infty})$$

Upper-Middle Nodes:
$$\frac{1}{4+2Bi+\frac{1}{Fo_c}} (T_{right} + T_{left} + 2T_{below} + \frac{1}{Fo_c} T_{node}^P + 2BiT_{\infty})$$

Upper-Right Corner Node:
$$\frac{1}{2+Bi+\frac{1}{2Fo_c}} (T_{left} + T_{below} + \frac{1}{2Fo_c} T_{node}^P + BiT_{\infty})$$

Left Wall Concrete Nodes:
$$\frac{1}{4+\frac{1}{Fo_c}} (T_{above} + 2T_{right} + T_{below} + \frac{1}{Fo_c} T_{node}^P)$$

Middle Concrete Nodes:
$$\frac{1}{4+\frac{1}{Fo_c}} (T_{above} + T_{left} + T_{right} + T_{below} + \frac{1}{Fo_c} T_{node}^P)$$

Right Wall Concrete Nodes:
$$\frac{1}{4+\frac{1}{Fo_c}} (T_{above} + 2T_{left} + T_{below} + \frac{1}{Fo_c} T_{node}^P)$$

Upper- Left Insulation Node:

$$\frac{1}{2+2\frac{k_i}{k_c}+\frac{1}{2Fo_c}+\frac{k_i}{k_c}\frac{1}{2Fo_i}} \left(T_{above} + \left(1 + \frac{k_i}{k_c} \right) T_{right} + \frac{k_i}{k_c} T_{below} + \left(\frac{1}{2Fo_c} + \frac{k_i}{k_c} \frac{1}{2Fo_i} \right) T_{node}^P \right)$$

Upper-Middle Insulation Nodes:

$$\frac{1}{4+4\frac{k_i}{k_c}+\frac{1}{Fo_c}+\frac{k_i}{k_c}\frac{1}{Fo_i}} \left(2T_{above} + \left(1 + \frac{k_i}{k_c} \right) T_{left} + \left(1 + \frac{k_i}{k_c} \right) T_{right} + 2\frac{k_i}{k_c} T_{below} + \left(\frac{1}{Fo_c} + \frac{k_i}{k_c} \frac{1}{Fo_i} \right) T_{node}^P \right)$$

Upper-Right Insulation Node:

$$\frac{1}{2+2\frac{k_i}{k_c}+\frac{1}{2Fo_c}+\frac{k_i}{k_c}\frac{1}{2Fo_i}} \left(T_{above} + \left(1 + \frac{k_i}{k_c} \right) T_{left} + \frac{k_i}{k_c} T_{below} + \left(\frac{1}{2Fo_c} + \frac{k_i}{k_c} \frac{1}{2Fo_i} \right) T_{node}^P \right)$$

Left Wall Insulation Nodes:
$$\frac{1}{4+\frac{1}{Fo_i}} (T_{above} + 2T_{right} + T_{below} + \frac{1}{Fo_i} T_{node}^P)$$

Middle Insulation Nodes:
$$\frac{1}{4 + \frac{1}{Fo_i}} (T_{above} + T_{left} + T_{right} + T_{below} + \frac{1}{Fo_i} T_{node}^P)$$

Right Wall Insulation Nodes:
$$\frac{1}{4 + \frac{1}{Fo_i}} (T_{above} + 2T_{left} + T_{below} + \frac{1}{Fo_i} T_{node}^P)$$

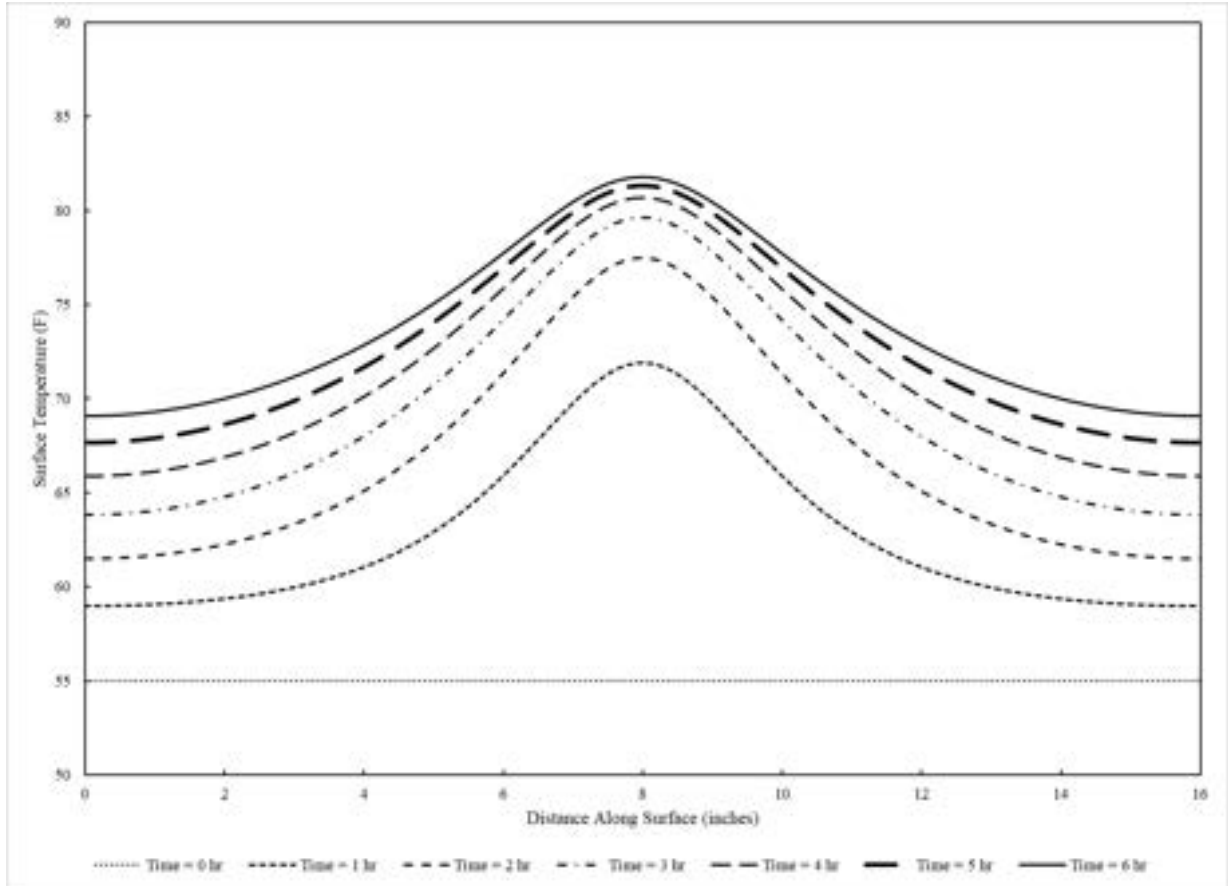


Figure 6. Transient model surface temperature during startup

The above figure shows the surface temperature distribution during startup, which occurs over a 6 hour period. The distribution approaches equilibrium as the average temperature approaches 95% of the steady state temperature.

This transient conduction model was used for finding the time required to reach 95% of steady state under various conditions. Using the model, we found that the initial startup time from a uniform temperature of 55°F was 6 hours. This felt reasonable, as the initial startup could be planned for the night before a period of time with relatively cold weather. After allowing the system to reach steady state, we tested the behavior after a 24 hour cooldown period. The temperature after this cooldown period was 67.8°F. After this cooldown period, it took only 3 hours to reach 95% of the steady state temperature. This felt like a very reasonable amount of time for a heating system to start up, so we decided that it was not necessary to specify a different startup condition.

Heat Transfer in Pipes

Table 5. Pipe Data. This table details the specifications for the pipes. Each value was calculated iteratively through the EES model.

Length of Each Tube	120 ft
Total Number of Loops	5
Flowrate Through Each Loop	$2085 \frac{lbm}{hr}$ {5.043 gpm}
Total Heat Transfer Rate	$15,738 \frac{Btu}{hr}$
Required Loop Inlet Temperature	93.73 °F
Temperature Drop Through Each Loop	1.3 °F
Final Pressure Drop in Each Loop	4.64 psi {10.76 ft head}

We will also compare this model to the ASHRAE Handbook which presents another method to design radiant heating systems. Figures 7 and 8 use the design graph for sensible heating and cooling with floor and ceiling panels. This figure, located in Chapter 6 of the 2016 ASHRAE Handbook-HVAC Systems and Equipment, is the method used by industry engineers to design radiant heating systems.

We performed two calculations using the panel design method, one using the surface heat flux and the other using the total heat flux with insulation included. First, the surface heat flux, total heat flux, panel thermal resistance, and tube spacing were calculated and tabulated in Table 6.

Table 6. ASHRAE Panel Design Method Data. $q_{u,surface}$ is the surface heat flux, $q_{u,total}$ is the total heat flux, r_u is the characteristic panel thermal resistance, and M is the tube spacing

$q_{u,surface}$	$16.53 \frac{Btu}{hr-ft^2}$
$q_{u,total}$	$19.67 \frac{Btu}{hr-ft^2}$
r_u	$0.3826 \frac{hr-ft^2-^{\circ}F}{Btu}$
M	16 in

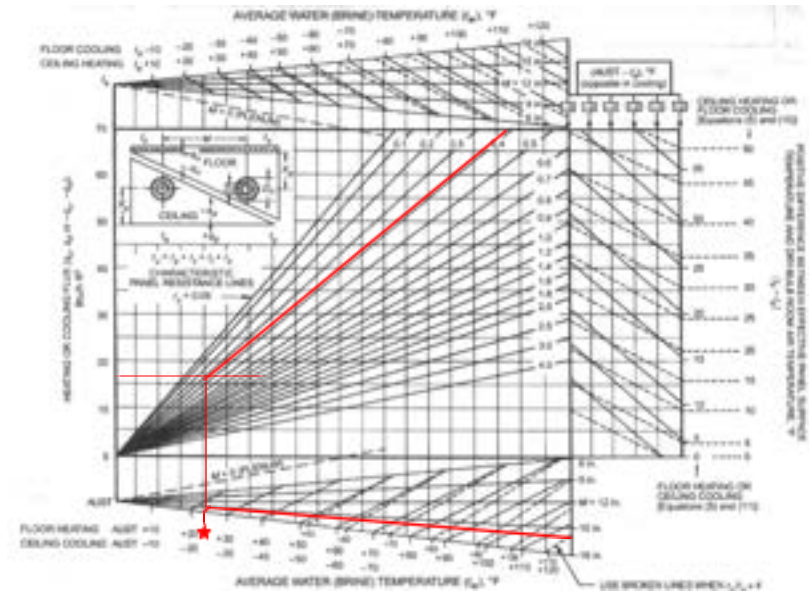


Figure 7: Panel Design Method Graph using the surface heat flux. Values from Table X are used.

From this method, the average water temperature t_w was $23^\circ\text{F} + 68^\circ\text{F} = 91^\circ\text{F}$. This is about 5°F less than our design average water temperature, T_m , of 96.26°F .

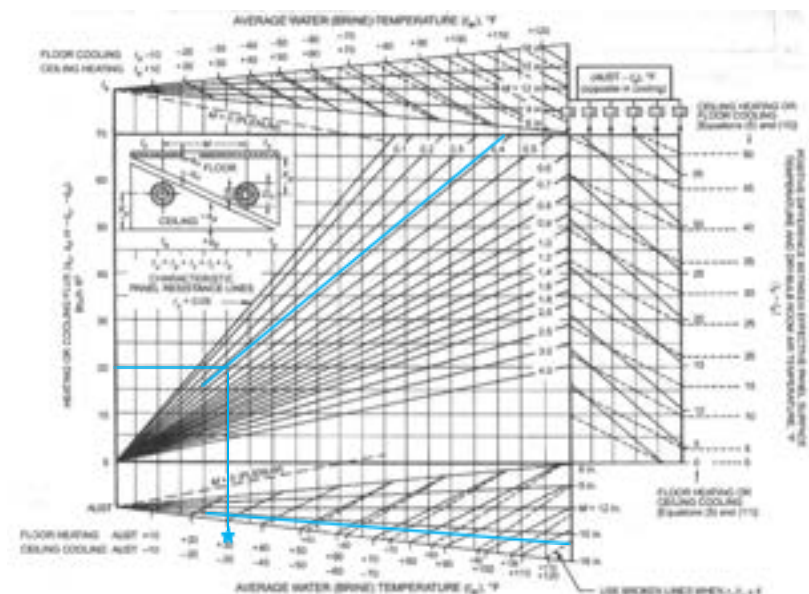


Figure 8: Panel Design Method Graph using the total heat flux. Values from Table X are used.

From this method, the average water temperature t_w was $30^\circ\text{F} + 68^\circ\text{F} = 98^\circ\text{F}$. This is about 2°F more than our design average water temperature, T_m , of 96.26°F .

Total System Pressure Drop and Pump Selection

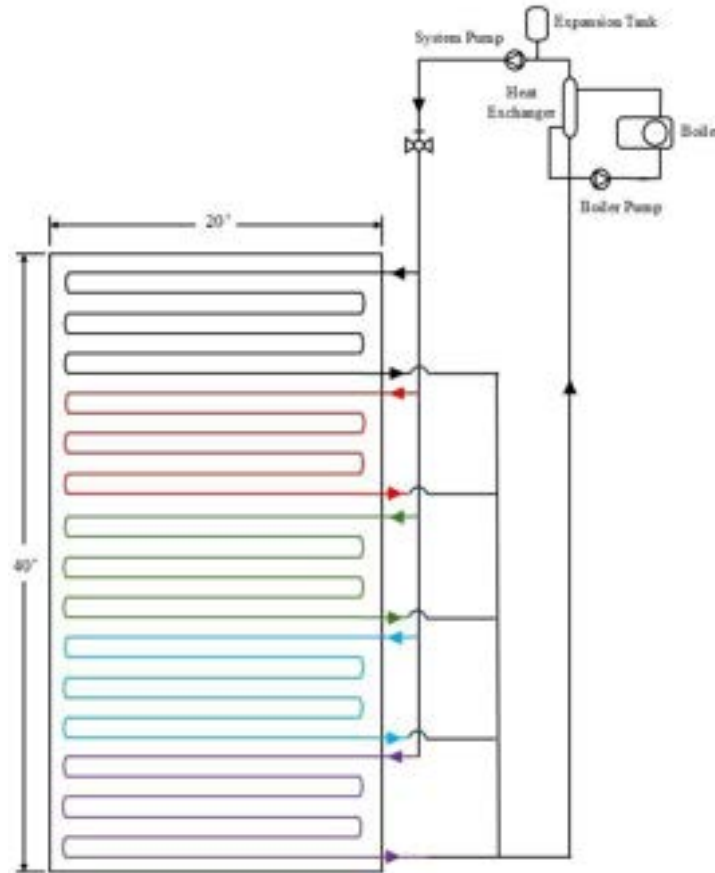


Figure 9: Schematic of Piping and Instrumentation Layout

Figure 9 shows the layout of the overall system. It shows 5 loops, each with 6 pipes. The use of reverse return piping allows for balancing valves to be omitted. Despite the additional piping cost, this layout will minimize operation and maintenance throughout the life of the system. The drawing is not to scale and is only to be used to communicate design intent, not for final installation.

The total pressure drop in the system is 28.22 ft, or 12.17 psi. This includes all fittings, piping, pump, and heat exchanger.

For our system, we selected the VersaFlo UPS32-160 Single Phase pump.

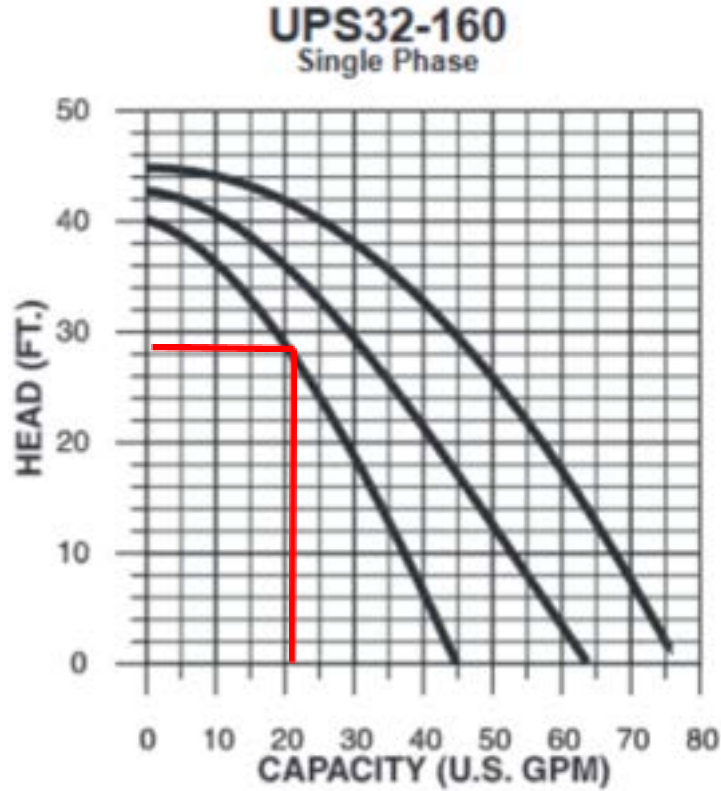


Figure 1: Performance curve for UPS32-160 pump from Grundfos. The intersection of the red lines is the operating point. Our system operates at approximately 21 gpm and 30 ft of head, making this an optimal pump choice.

This pump will operate at 21 gpm and 30 ft of head. The BHP is 0.25 hp with a 60% pump efficiency.

Heat Exchanger Selection

An expansion tank is necessary for any closed system.

We first found the total volume of fluid in the system using the following relationship:

$$V_{fluid} = \frac{\pi}{4} D_{loop}^2 L_{loop} + \frac{\pi}{4} D_{PVC}^2 L_{PVC}$$

D_{loop} and L_{loop} are the inner diameter and total length of the PEX heating tubes. D_{PVC} and L_{PVC} are the inner diameter and total length of the PVC pipes.

Using ideal gas relationships we found:

$$V_{tank} = \frac{V_{fluid} \left(\frac{v_2}{v_1} - 1 \right)}{1 - \frac{p_1}{p_2}} = 0.1048 \text{ ft}^3, \text{ or } 0.7842 \text{ gal}$$

Where v_1 is the specific volume of water at the minimum expected temperature of 55°F and v_2 is the specific volume of water at the maximum expected temperature of 140°F. P_1 is the minimum pressure at the low temperature, which we chose to be 14.7 psia in order to be above the manufacturer's NPSHR of 11.6 psia. P_2 is the maximum system pressure at the high temperature, which we chose to be 60 psia.

With our design expansion tank of 0.7842 gallons, we selected the Watts 2.1 Gallon Potable Water Expansion Tank.

Next, we selected the SEC PL/PLT 45 Shell & Tube Heat Exchanger. The performance data for this heat exchanger is shown in Table 7.

Table 7. Manufacturer Data for the SEC PL/PLT 45 Shell & Tube Heat Exchanger.

Nominal Capacity		Hot water				Cold Water				Heat Transfer Surface	
		Flow		Pressure Drop		Flow		Pressure Drop			
kW	Btu/hr	l/min	gpm	kPa	psig	l/min	gpm	kPa	psig	m²	ft²
13	45,000	23	6.1	6.2	0.90	150	39.6	7.4	1.07	0.150	1.62

Using the manufacturer's data, the overall conductance (UA) was calculated to be 454.4 Btu/hr-°F. At both steady state and startup, the heating loop will be operated at 5% flow.

Using the built in EES correction factor function with our design inlet and outlet temperatures for both sides of the heat exchanger, we found the correction factor to be 0.9716. Since this is above 0.8, we deem this an adequate design.

Cost Estimation

The total upfront cost of installing our system in the community center atrium was estimated to be \$16,397.44. The additional cost of electricity and gas was estimated to be \$289.01/yr based on an estimated heating load of 8 hours/day for 6 months, with a correction factor of 50% to account for the varying heat loads throughout the cold season. Accounting for a 6% interest rate, the total present value of the system over a 15-year life is \$19,204.38.

For comparison, we analyzed the cost of a similar electric radiant heating system, which costs \$20/ft² installed (without insulation), and found that the total present cost of the competing system was \$31,772.80 after accounting for insulation and energy cost. This cost is significantly higher than that of the proposed hydronic radiant heating system, even after accounting for the approximately \$3,000 spent on engineering hours.

Further development of the system will include considerations for system control methods, including thermostat control, a control valve in the heating loop, and potentially controlling the boiler to further reduce energy consumption. This will require upfront costs for materials, installation, and programming, as well as future maintenance and electricity costs.

Conclusion

The radiant heating system designed for the Atascadero community center atrium is an efficient and cost-effective solution that meets all design requirements. The five-loop design with $\frac{3}{4}$ -inch PEX tubing at 16-inch spacing provides a reasonable balance between performance and economy with 13,224 Btu/hr of heating capacity and startup times of 3-6 hours.

Our economic analysis demonstrates significant cost advantages over comparable electric radiant systems, with a present value of \$19,204.38 over a 15-year lifespan compared to \$31,772.80 for the alternative solution. The Grundfos UPS32-160 pump and SEC PL/PLT 45 heat exchanger ensure reliable operation with appropriate safety margins. This hydronic radiant heating system will provide comfortable, energy-efficient heating for the atrium.